

Hulle_Madeline_Individual_Assignment_05

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2026-05-01

Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#survey data (reading different sheets in as seperate data frames)

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information

sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quaity

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # values in df that are actually indicating missing values,
  # making it seem as a chr when it should be num
  # different ways missing values show up in data set
  na = c("over", "", "n/a", "N/A", "173+")
)

```

Class Code

2. Cleaning

sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sites
  filter(sector == "NCOS" &
    sampling_frequency == "Quarterly")

```

clean taxon_list

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean water_quality

```
# creatign new object from water_quality
water_quality_clean <- water_quality |>
  # cleaning column name
  clean_names() |>
  # a filtering join, only keep the rows in wq clean in which the
  # site matches up with the site in ncos clean
  semi_join(ncos_sites, by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating the medians
  summarize(
    med_pH = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping df
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```
# create a new object
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join only include quarterly ncos surveys
  semi_join(ncos_sites, by = "site") |>
```

```

# renaming date_on_vial column to date
rename(date = date_on_vial) |>
# selecting columns of interest
# keeping information about each survey and the 9 most abundant taxa
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filtered to only include filteres beaker samples
# (better sampling technique)
filter(sample_type %in% c("FB 250", "FB250")) |>
# converting to long format, making df longer
pivot_longer(
  # name the columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for old column names
  # column names are all in new column called taxon
  names_to = "taxon",
  # new column for the values in each of the old columns
  # counts now under a column called abundance
  values_to = "abundance"
) |>
# group by date site and taxon
group_by(date, site, taxon) |>
# calculated average count per liter (total abundance/7.7)
summarise(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungrouping data frame
ungroup() |>
# new column to join data frames, didnt remove them
unite("date_site", date, site, remove = FALSE) |>
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(

```

```

site == "NVBR" ~ "North of Venoco Bridge",
site == "NEC" ~ "East Channel",
site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>
# set factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                              med_sal,
                              .fun = "median",
                              .na_rm = TRUE),
       # reversing the order
       site_full = fct_rev(site_full))

```

Visualizing

Salinity

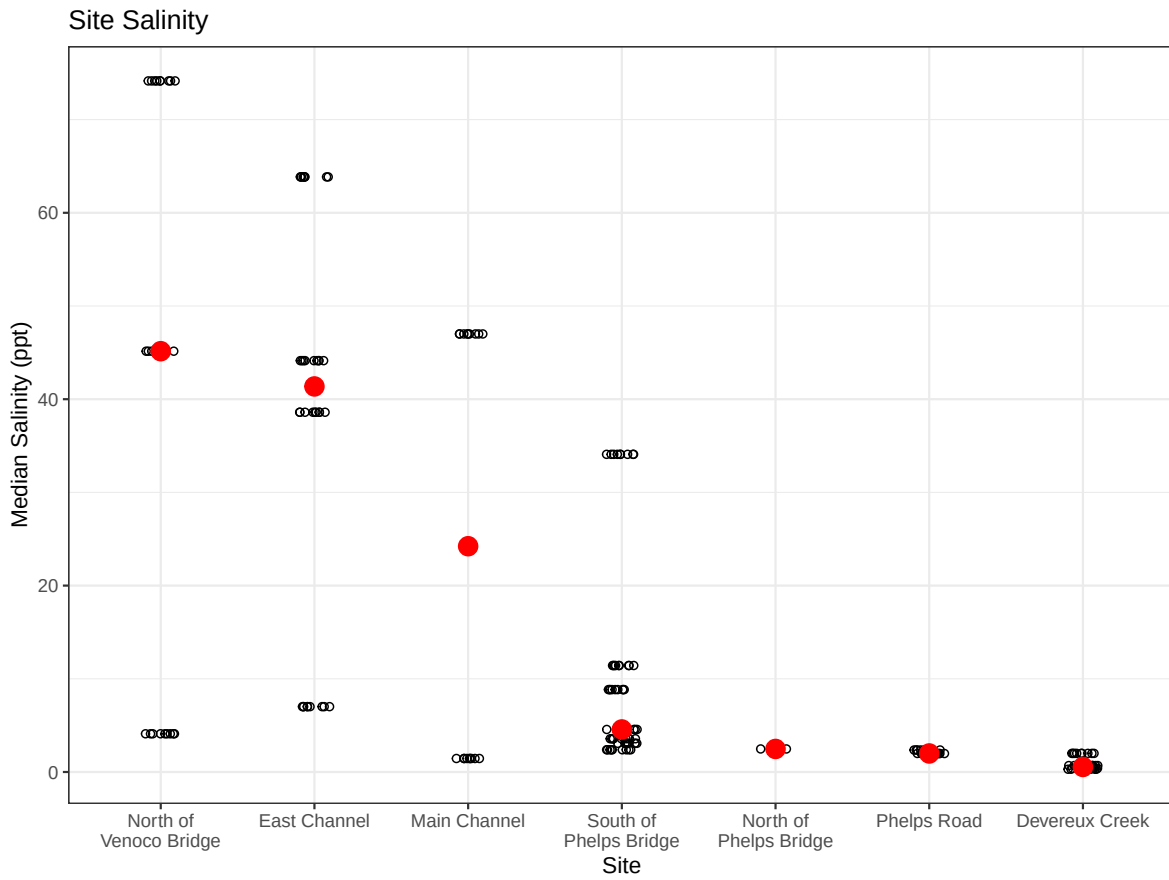
```

# creating new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
# points representing median salinity at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points representing mean salinity
stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 4) +
# adding line breaks to x axis labels (wrapping)
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes and adding a title
labs(x = "Site",
     y = "Median Salinity (ppt)",
     title = "Site Salinity") +

```

```
# adding themeing
theme_bw()

# visualizing the object
salinity_plot
```



Copepod Abundance

Cleaning

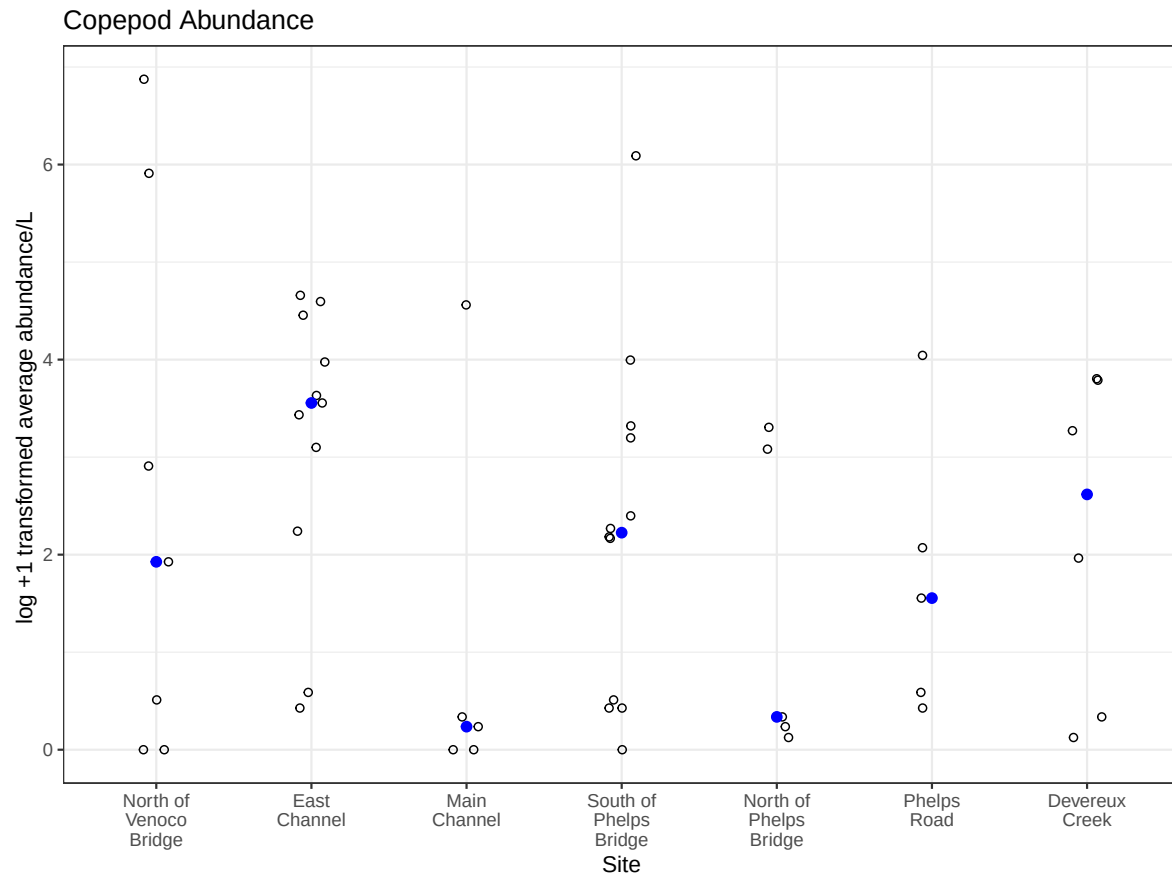
```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # adding 1 bc some observations are 0
  # helps overcome issues with large differences in order of magnitude
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod Viz

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    # using log transformed count per liter
                                    y = log_ave_lit)) +

# jitter to represent each sample
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points to represent median copepod abundance
stat_summary(geom = "point",
            fun = median,
            color = "blue",
            size = 2) +
# wrappign x-axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +
# relabeling axes
labs(x = "Site",
     y = "log +1 transformed average abundance/L ",
     title = "Copepod Abundance") +
# using different theme
theme_bw()

# displaying the object
copepod_plot
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit` (updated w/ `log_ave_lit` after viz)
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary(geom = "point", fun = median)`

b. Wrangle your data

```
# creating a new object `ostracods` from `aq_ins_clean`
ostracods <- aq_ins_clean |>
  # filtering to only include ostracods
  filter(taxon == "ostracod") |>
  # log transforming `ave_lit` to overcome large difference in
  # order of magnitude between observations
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a sample of 10 observations
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
  date_site      date            site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <chr>  <dbl>  <dbl>  <dbl>  <dbl>
1 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1 ostr~ 0      NA      NA      NA
2 2021-11-16_NPB 2021-11-16 00:00:00 NPB  ostr~ 72.1    7.3    2.38   12.1
3 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR ostr~ 0      8.86   45.2   10.2
4 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2 ostr~ 0.133   7.71    1.98   9.59
5 2022-05-06_NMC 2022-05-06 00:00:00 NMC  ostr~ 0      9.96    1.45  16.5
6 2023-11-11_NEC 2023-11-11 00:00:00 NEC  ostr~ 0      NA     38.6    0.25
7 2023-09-15_NMC 2023-09-15 00:00:00 NMC  ostr~ 0.4    NA     NA     14.2
8 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR ostr~ 0      NA     NA     NA
9 2022-01-28_NDC 2022-01-28 00:00:00 NDC  ostr~ 9.47    7.65    2.00   9.55
10 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2 ostr~ 2      7.68    2.36   2.43
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code Your Figure

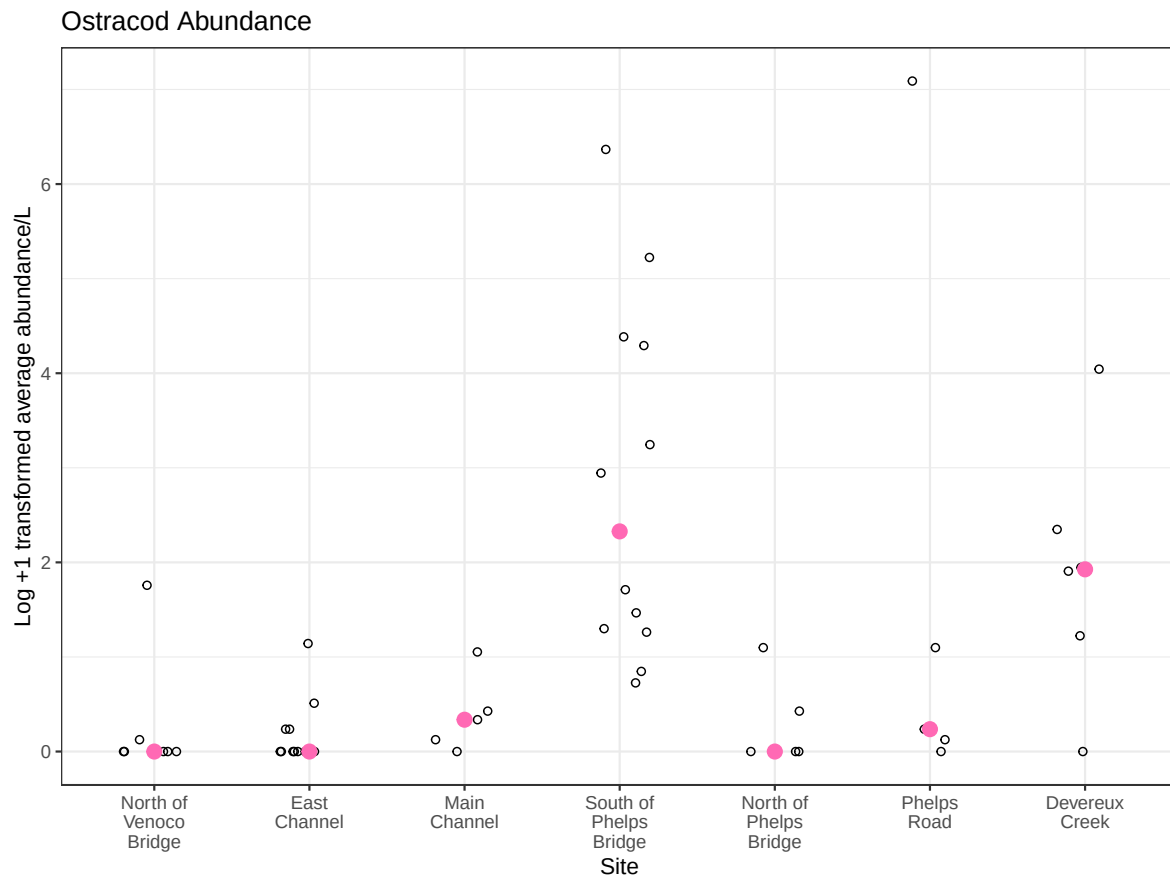
```
ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  geom_jitter(height = 0,
              width = 0.2,
              shape = 21) +
```

```

stat_summary(geom = "point",
             fun = median,
             color = "hotpink",
             size = 3) +
labs(x = "Site",
     y = "Log +1 transformed average abundance/L",
     title = "Ostracod Abundance") +
theme_bw() +
scale_x_discrete(labels = label_wrap(width = 10))

ostracod_plot

```



d. Create a Multi Panel Figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```

